

OSINT LAB – RECON NG

CYBERSECURITY TECHNOLOGIES AND RISK MANAGEMENT –
A.Y. 24/25



Università degli Studi di Cagliari

OSINT LAB – RECON-NG

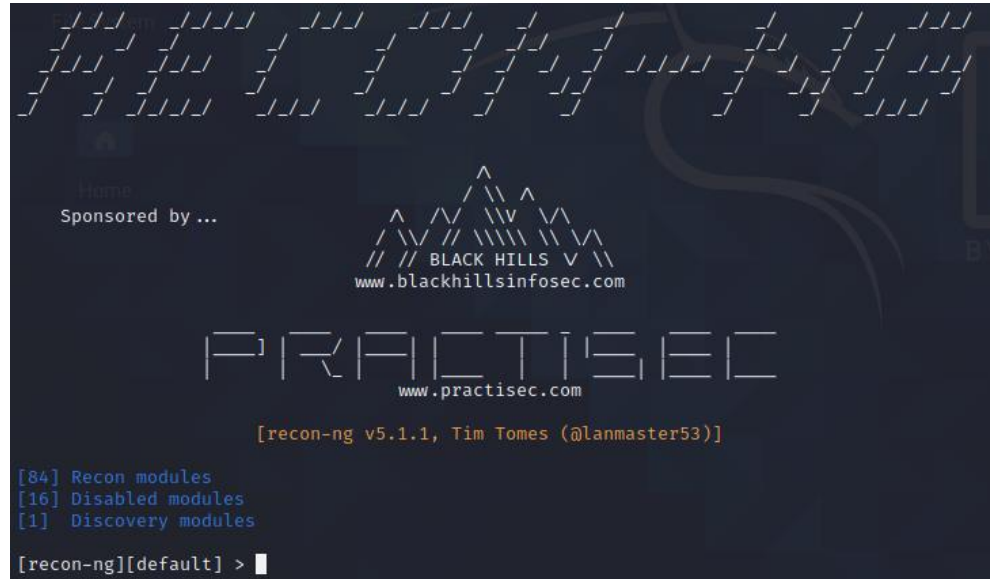
The easiest way to use **recon-ng** for the OSINT Lab and the OSINT homework is to install a Kali Linux Virtual Machine on your computer with either:

1. **VirtualBox**
2. **VMWare**

RECON NG

Recon-ng is a full-featured reconnaissance framework designed with the goal of providing a powerful environment to conduct open source web-based reconnaissance quickly and thoroughly.

You can run the tool by clicking on the Menu > Information Gathering > Recon on the Linux GUI, or by typing **recon-ng** on a terminal emulator.



RECON NG - COMMANDS

Most useful recon-ng commands:

help – shows the list of all commands and their functionalities

db – interfaces with the database

marketplace – interfaces with the marketplace, where modules can be downloaded

modules – used to search, load or reload modules

workspaces – manages the workspaces

It is possible to type any command and then type ENTER to see the correct usage.

RECON NG - DATABASE

Data in Recon-ng is organized in tables.

Every table has a pre-defined name, and each row of the table is identified by an ID (rowid).

Use the command **db schema** to see the database scheme (names of the tables, names and domains of their columns).

```
[recon-ng][default] > db schema
```

domains	
domain	TEXT
notes	TEXT
module	TEXT

companies	
company	TEXT
description	TEXT
notes	TEXT
module	TEXT

netblocks	
netblock	TEXT
notes	TEXT
module	TEXT

RECON NG - WORKSPACES

Different working sessions can be associated with different workspaces. Each workspace has its own database and activity.

Use the command **workspaces create <name>** to create a new workspace.

Using **workspaces load <name>** you can switch between workspaces.

The command **workspaces list** lists all the workspaces.

```
[recon-ng][test1] > workspaces list
```

Workspaces	Modified
default	2021-04-26 17:19:40
test	2021-04-26 18:27:50
test1	2021-04-29 12:17:44

RECON NG - MODULES

Recon-ng is based on modules, i.e., pieces of python code that, given some inputs, retrieve data from open sources.

When a module is run, retrieved data is stored on the relevant database tables.

The command **marketplace search** shows the list of all available modules.

To install a module, run the command **marketplace install <name of the module>/**

For this Lab, three modules are needed (they are already installed in your Azure Labs VM):

recon

discovery

reporting

RECON NG - MODULES

To use a module, first load it by typing **modules load <name of the module>**. After loading a module, you can type **info** to get a description of its functionality, the database tables where the results will be stored, and its options.

```
[recon-ng][test1] > modules load whois_pocs
[recon-ng][test1][whois_pocs] > info

    Name: Whois POC Harvester
    Author: Tim Tomes (@lanmaster53)
    Version: 1.0

Description:
    Uses the ARIN Whois RWS to harvest POC data from whois queries for the given domain. Updates the
    'contacts' table with the results.

Options:


| Name   | Current Value | Required | Description                              |
|--------|---------------|----------|------------------------------------------|
| SOURCE | default       | yes      | source of input (see 'info' for details) |



Source Options:
    default      SELECT DISTINCT domain FROM domains WHERE domain IS NOT NULL
    <string>      string representing a single input
    <path>        path to a file containing a list of inputs
    query <sql>  database query returning one column of inputs
```


RECON NG - MODULES

Before running a module, some options can be set.

For instance, to set the SOURCE option in the whois_pocs module, type the command **options set SOURCE <source>**

Once the options are set, the module can be run with the command **run**

The results will be automatically stored on the relevant tables. The content of a table can be displayed with the command **show <table name>**

```
[recon-ng][test1][whois_pocs] > options set SOURCE microsoft.com
SOURCE ⇒ microsoft.com
[recon-ng][test1][whois_pocs] > run

_____
MICROSOFT.COM
_____
[*] URL: http://whois.arin.net/rest/pocs;domain=microsoft.com
[*] URL: http://whois.arin.net/rest/poc/AADLA11-ARIN
[*] Country: United States
[*] Email: v-chrisa@microsoft.com
[*] First_Name: CHRIS
[*] Last_Name: AADLAND
[*] Middle_Name: None
[*] Notes: None
[*] Phone: None
[*] Region: Seattle, WA
[*] Title: Whois contact
[*] _____
[*] URL: http://whois.arin.net/rest/poc/AADLA1-ARIN
```

RECON NG – INTERESTING MODULES

The following modules don't require any API key and are quite powerful:

- **bing_domain_web** ? finds hosts from a domain
- **brute_hosts** ? finds hosts from a domain, more powerful
- **brute_suffix** ? finds subdomains from a domain
- **resolve** ? finds hosts' IP addresses
- **reverse_resolve** ? finds hosts associated to an IP address
- **whois_miner** ? finds anything about a company, powerful
- **whois_pocs** ? finds contacts from a domain
- **interesting_files** ? finds interesting files from hosts in predictable locations

RECON NG – REPORTING

Recon-ng has specific modules to create reports in a quick way and in several formats:

- **csv**
- **html**
- **json**
- **list**
- **xlsx**
- **xml**
- ...

These modules automatically generate a report in the corresponding format, and store it in a predefined directory. The absolute filename is shown as the output of the **run** command.

Whois & ICANN Lookup

Find all domain information related to the IP or website

<https://whois.domaintools.com/>

<https://lookup.icann.org/en>

Retrieve data of the registered domain



DomainTools



ICANN

Whois

Whois Record for Unica.it

Domain Profile

Registrar Status	ok	
Dates	10,277 days old Created on 1996-01-29 Expires on 2025-01-29 Updated on 2024-02-14	↗
IP Address	130.186.7.245 - 11 other sites hosted on this server	↗
IP Location	 - Bologna - Ceretolo - Cineca Consorzio Interuniversitario	
ASN	 AS198949 Radware Radware Ltd, IL (registered Jun 28, 2012)	
IP History	5 changes on 5 unique IP addresses over 1 years	↗

Whois Record (last updated on 2024-03-19)

```
Domain Name: unica.it
Registry Domain ID:
Registrar WHOIS Server:
Registrar URL:
Updated Date: 2024-02-14T01:02:44
                2017-09-19 16:07:14
                2017-09-19 16:13:54
                2023-11-24 16:55:50
                2017-09-19 16:17:27
                2017-09-20 09:15:37
                2024-02-14
Creation Date: 1996-01-29T00:00:00
                2007-03-01 10:47:03
                2013-12-03 15:14:13
                2015-07-14 18:23:57
                2013-12-03 15:15:21
                2015-07-14 18:27:06
                1996-01-29
Registrar Registration Expiration Date: 2025-01-29T00:00:00
Registrar:
Sponsoring Registrar IANA ID:
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Status:
        ok
```

Whois

Registry Registrant ID:
Registrant Name:
Registrant Organization:
Registrant Street:
Registrant City:
Registrant State/Province:
Registrant Postal Code:
Registrant Country:
Registrant Phone:
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email:
Registry Admin ID:

Admin Name:
Admin Organization:
Admin Street:
Admin City:
Admin State/Province:
Admin Postal Code:
Admin Country:
Admin Phone:
Admin Phone Ext:
Admin Fax:
Admin Fax Ext:
Admin Email:

Tech Name:
Tech Organization:
Tech Street:
Tech City:
Tech State/Province:
Tech Postal Code:
Tech Country:
Tech Phone:
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email:

Billing Name:
Billing Organization:
Billing Street:
Billing City:
Billing State/Province:
Billing Postal Code:
Billing Country:
Billing Phone:
Billing Phone Ext:
Billing Fax:
Billing Fax Ext:
Billing Email:
Nameservers:

DNSSEC: no

ICANN Lookup

Convert website address to IP address: <https://domaintoipconverter.com/www.unica.it> -> 90.147.144.44

Enter a domain name or an Internet number resource (IP Network or
ASN)

[Frequently Asked Questions](#)
(FAQ)

Lookup

By submitting any personal data, I acknowledge and agree that the personal data submitted by me will be processed in accordance with the ICANN [Privacy Policy](#), and agree to abide by the website [Terms of Service](#) and the [registration data lookup tool Terms of Use](#).

ICANN Lookup

IP Network Information

Handle: 90.147.144.0 - 90.147.147.255

Status:
active

Address Range: 90.147.144.0 - 90.147.147.255

IP version: v4

Name: Uni-Cagliari-IPv4-NET

Type: ASSIGNED PA

Country Code: IT

Parent Handle: 90.147.0.0 - 90.147.255.255

Whois Server: whois.ripe.net

Dates

Registration: 2014-01-16 13:57:36 UTC

Last changed: 2022-01-14 19:18:42 UTC

Address range assigned to the domain

Name of the network

Country code where the domain is registered

ICANN Lookup

Contact Information

Registrant:

Handle: GARR-LIR

Registrant:

Handle: ORG-UDSD50-RIPE

Administrative:

Handle: RP5141-RIPE

Technical:

Handle: SM19296-RIPE

Technical:

Handle: VE853-RIPE

The contact information of the registrant, administrative, technical and billing contacts

Abuse:

Handle: AG16225-RIPE

Name: ABUSE GARR

Email: lir@garr.it

Kind: group

Mailing Address: Consortium GARR, Via dei Tizii, 6, I-00185 Roma, Italy

Registrant:

Handle: GARR-LIR

Technical:

Handle: GL965-RIPE

Administrative:

Handle: LL8984-RIPE

ICANN Lookup

Authoritative Servers

Registry Server URL: <https://rdap.db.ripe.net/ip/90.147.144.44>

The URL(s) of the authoritative servers (sources) of the lookup results

Notices and Remarks

Remarks:

UNI-Cagliari

Notices:

Filtered

This output has been filtered.

Whois Inaccuracy Reporting

If you see inaccuracies in the results, please visit:

https://www.ripe.net/contact-form?topic=ripe_dbm&show_form=true

Source

Objects returned came from source

RIPE

Terms and Conditions

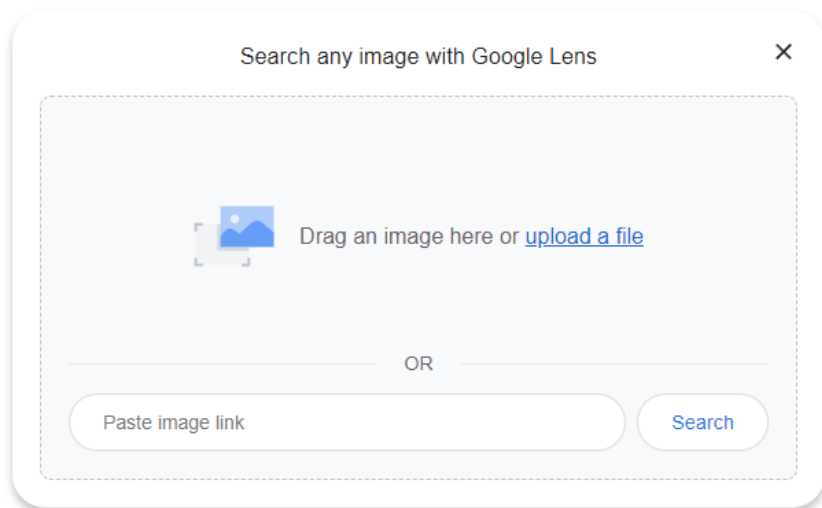
This is the RIPE Database query service. The objects are in RDAP format.

<http://www.ripe.net/db/support/db-terms-conditions.pdf>

Google Image - Search

This very powerful functionality can be used in many ways

- **Related images**
- **Websites that are using the image**
- **Geolocating landscapes**



Google Image

Search. The functionality of Google to search for **related images, websites** containing the image, and **geolocalize** landscapes

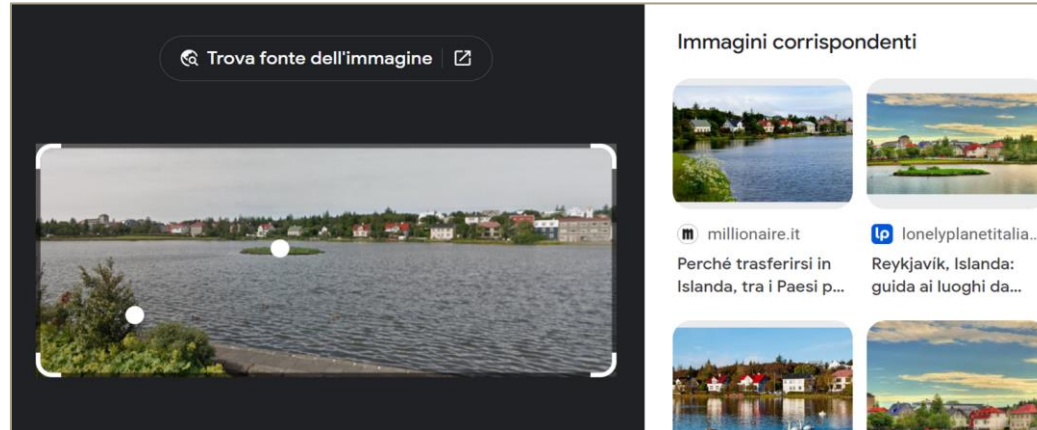
Example. Were are we?



Google Maps - Street View

Example. Where are we?

Step 1. Related images to find that we are in Reykjavik (Iceland)

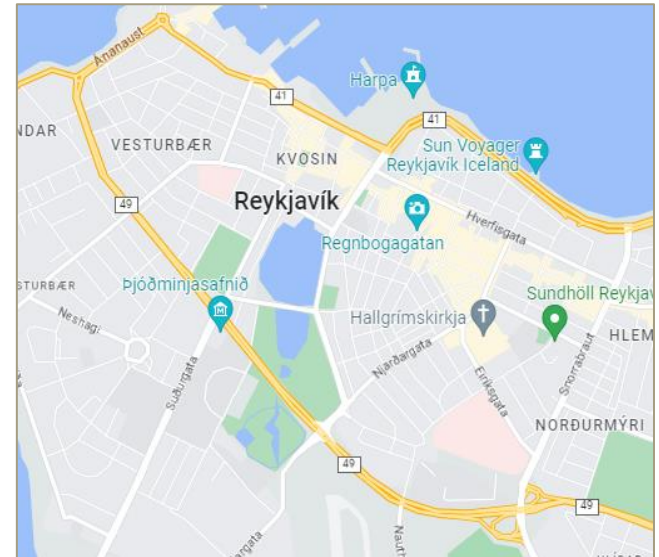


Google Maps - Street View

Example. Where are we?

Step 1. Related images to find that we are in Reykjavik (Iceland)

Step 2. Satellite View of Reykjavik on Google Maps to see that there is a lake



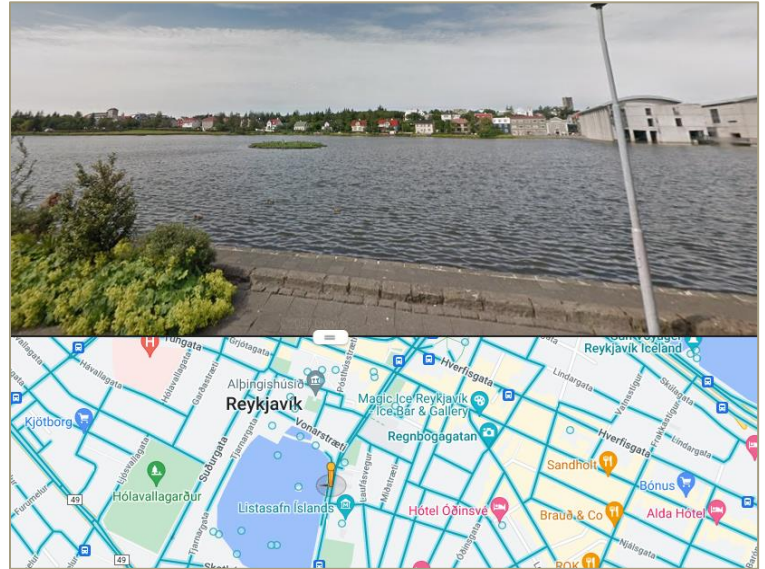
Google Maps - Street View

Example. Where are we?

Step 1. Related images to find that we are in Reykjavik (Iceland)

Step 2. Satellite View of Reykjavik on Google Maps to see that there is a lake

Step 3. See the images of the lake on Google Maps Street View



Internet Archive

The internet archive (<https://archive.org/web/>) is a website that contains snapshots of web pages taken during time



Explore more than 802 billion [web pages](#) saved over time

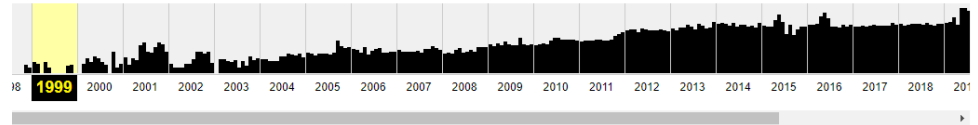
BROWSE HISTORY

Internet Archive

It allows to retrieve very old versions of popular websites

Most importantly it allows to retrieve contents that were deleted

- **Social media posts**
- **Websites**



Search the web using Google!

Google Search

I'm feeling lucky

Sherlock

A tool to retrieve all social network links associated to a given username

<https://github.com/sherlock-project/sherlock>

Very useful for:

- unknown social network/websites
- automated search
- quick scan in popular and unknown socials or limit to a set
- browse to all results to default browser



```
~/sherlock
$ python3 sherlock hackerman1337
[*] Checking username hackerman1337 on:
```

Sherlock

Usage example: username ctrm_osint

```
(kali㉿kali) - [~]  
$ sherlock ctrm_osint  
[*] Checking username ctrm_osint on:  
  
[+] Coil: https://coil.com/u/ctrm_osint  
[+] TikTok: https://tiktok.com/@ctrm_osint  
[+] igromania: http://forum.igromania.ru/member.php?username=ctrm_osint  
[+] jeuxvideo: http://www.jeuxvideo.com/profil/ctrm_osint?mode=infos  
  
[*] Search completed with 4 results
```

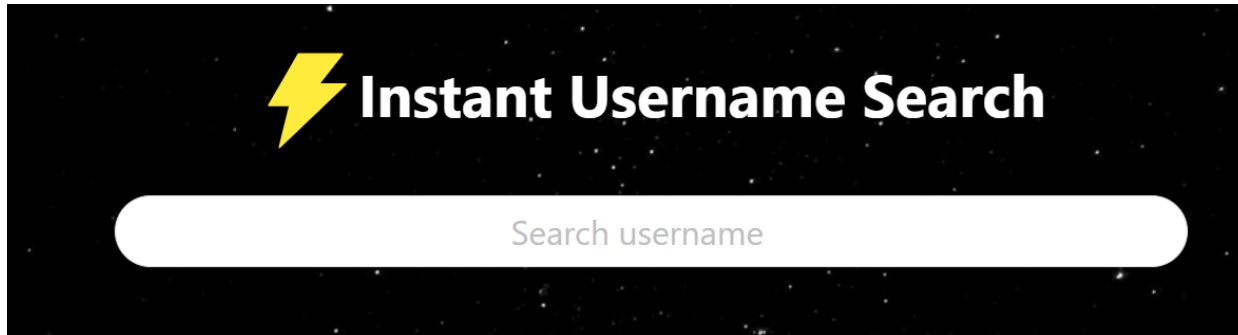
Instant Username Search

Website alternative to Sherlock: search the username in 100 social networks

<https://instantusername.com/#/>

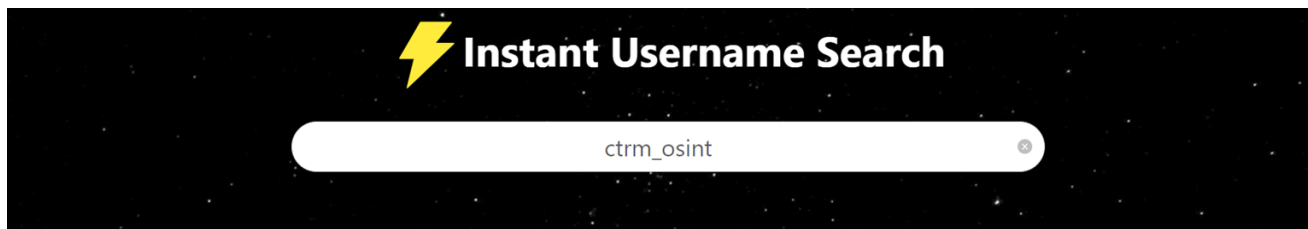
Useful not only to search a user but also to check if the username has already been taken in a specific social network and so create a new account

Solve OSINT username research even without a computer and Kali Linux distribution



Instant Username Search

Usage example: username ctrm_osint



Instagram Available	TikTok Available	Twitter Available	Facebook Taken	YouTube Available
Medium Taken	Reddit Available	HackerNews Available	GitHub Available	Quora Available
9GAG Available	VK Available	GoodReads Available	Blogger Taken	Patreon Available

CHALLENGE 1

Retrieve the GPS coordinates of [this picture](#)



CHALLENGE 2

Retrieve the name of this river



CHALLENGES 3

Retrieve the date when this image has been taken



CHALLENGE 4

Retrieve the secret from [this webpage](#)
Hint: look for something of summer/autumn 2022



CHALLENGE 5

Use recon-ng to find all contacts of unica.it



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A. Kali Linux Installation

Step 1. Download and install Virtual Box (or VMware)

Step 2. Download the prebuilt VM from

<https://www.kali.org/get-kali/#kali-virtual-machines>

for Vbox or VMware

Step 3. Extract the content of the zipped file and double-click on the .vbox (or open the .vmx file with Vmware)

Step 4. Double-click on the virtual machine just created and login with
username: kali
password: kali

NB. If you need to change the settings of the VM (RAM, disk, etc.) you can do that before switching on the VM

A. Kali Linux Installation

